

BI Project Report

INTERNATIONAL CONSORTIUM FOR MEDICAL RESEARCH ADVANCEMENT (IMCRA)

STRATEGIC ANALYTICS & VISUALISATION REPORT

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1. Executive Summary

This report presents a strategic analysis of ICMRA's fundraising operations across 2019 to 2025, covering \$32.93 million in total pledges from 2,330 unique donors across 45,100 transactions. ICMRA achieved a campaign return on investment (ROI) of 1,077%, raising \$11.78 for every dollar spent on campaign budgets. However, the data reveals a critical inflection point: after four consecutive years of triple-digit growth, revenue declined by 17.3% in 2024 and a further 27.5% in 2025. Six strategic insights are presented, each grounded in advanced analytical techniques including RFM segmentation, customer lifetime value modelling, market basket analysis, scenario planning, and cohort retention tracking. The evidence points to three systemic risks: geographic concentration (71% of revenue from the United States), campaign concentration (50.3% from a single campaign), and donor type homogeneity (94.9% from households). These risks, if unaddressed, threaten ICMRA's long-term fundraising sustainability.

2. Data Quality & Assurance

2.1 Data Sources and Scope

The analysis draws on four primary tables sourced from the ICMRA Dataset 2019-2025 workbook: the Account dimension (2,330 records), the Medical Research Grant Appeal fact table (45,100 transaction records), the MRA Campaign dimension (22 campaigns), and the Contribution type lookup (10 contribution categories). The dataset spans seven calendar years (January 2019 to December 2025).

2.2 Quality Assessment

A systematic data quality assessment was conducted during the Power Query ingestion phase. The dataset was found to be structurally complete with no missing values in critical fields. All 45,100 records contain valid Account IDs, Campaign IDs, Contribution Type IDs, and monetary amounts. Date fields (Appeal Close Date and Payment Date) are fully populated with no null entries. Monetary values in the Pledge-Pay Amount column are consistently positive with no negative or zero-dollar anomalies detected.

Minor data standardisation issues were identified and resolved: five Account records carried the account type label "Househld" rather than the standard "Household" classification, representing \$72,357 (0.2% of total revenue). These were corrected via a Power Query value replacement step to ensure consistent grouping. Campaign budget and target fields were validated against the Campaign dimension with no discrepancies found. All foreign key relationships between the fact table and dimension tables were verified as referentially complete; every transaction references a valid Account, Campaign, and Contribution Type.

2.3 Transformation Summary

All data cleaning was performed exclusively within the Power Query engine. Applied steps include: column type enforcement (text, integer, datetime), header promotion, the aforementioned value replacement for account type standardisation, and the generation of a custom date dimension table (Dim_Date) to support time-intelligence calculations. The complete step-by-step transformation log is documented in Appendix B.

See Appendix A for the detailed data quality log and Appendix B for Power Query transformation documentation.

3. Insight 1: Revenue Trajectory: Hypergrowth, Peak, and Decline

The Growth Story (2019 to 2023)

ICMRA experienced extraordinary fundraising growth from 2019 to 2023. Revenue expanded from \$820,998 to \$8,058,372, a compound annual growth rate (CAGR) of 77%. Year-over-year growth was consistently exceptional: +102% (2020), +108% (2021), +115% (2022), and +8.4% (2023). This was driven by donor acquisition, increased transaction frequency (averaging 19.3 per donor), and the scaling of high-performing campaigns, particularly Program Funding at \$25.97 million across seven campaigns (2,310% ROI).

The Inflection Point (2024 to 2025)

The growth trajectory reversed sharply. Revenue fell to \$6,660,515 in 2024 (−17.3%) and further to \$4,832,165 in 2025 (−27.5%), a 40% decline from the 2023 peak. Several structural factors may explain this. First, growth deceleration was already visible in 2023 (+8.4% versus +115% in 2022), suggesting organic market saturation. Second, the top-performing campaign (Cam-Res-App-242) accounts for 50.3% of total revenue at \$16.56 million; any underperformance disproportionately affects aggregate results. Third, monthly seasonality reveals heavy reliance on Q4 giving (29.8% of annual revenue), with December alone contributing 14.4%.

Revenue is back-loaded: Q1 (25.9%), Q2 (22.8%), Q3 (21.6%), and Q4 (29.8%), with December alone contributing 14.4%. Mid-year months such as May (5.5%) and July (5.8%) are consistently weaker, suggesting a seasonal trough during the northern hemisphere summer. With 70.7% of all revenue originating from the United States, macroeconomic factors affecting US household giving could explain a significant portion of the downturn. The decline accelerated from −17% to −28%, suggesting a compounding effect rather than a one-time adjustment.



Figure. Executive Overview page

4. Insight 2: RFM Segmentation: The Champion-Loyal Donor Engine

Methodology

An RFM (Recency, Frequency, Monetary) analysis was conducted to segment ICMRA's 2,330 unique donors based on the recency of their last pledge, the frequency of their transactions, and the total monetary value of their contributions. Each donor was assigned a composite score on a 1 to 5 scale across all three dimensions using PERCENTILE.EXC-based bucketing within the Power BI semantic model. These scores were then mapped to segment labels through the RankRFM lookup table.

Segment Distribution

The segmentation reveals a donor base that is remarkably healthy by industry standards. Champions, donors with high recency, high frequency, and high monetary scores, number 314 (13.5% of donors) and contribute \$10.81 million, or 32.8% of total revenue. The Loyal segment, the largest at 1,623 donors (69.7%), contributes \$19.12 million (58.1%). Together, Champions and Loyal donors account for 90.9% of all revenue. The Potential Loyalist segment contains 334 donors

contributing \$2.65 million (8.0%), while the At Risk segment (49 donors, \$313,107) and Lost segment (10 donors, \$27,169) are negligible at a combined 1.1%.

Strategic Implications

The concentration of revenue in the Champion and Loyal segments is both a strength and a vulnerability. On one hand, the near-zero churn rate (0.4%) and the dominance of repeat donors (99.8% of donors contribute to multiple campaigns, with an average of 5.05 campaigns per donor) indicates deep donor engagement. On the other hand, the low number of Champions (314) means that disengagement of even a small number could materially impact revenue. The Potential Loyalist segment (334 donors) represents the most actionable growth opportunity: targeted stewardship programmes could convert these donors into Champions, potentially unlocking significant incremental revenue. Khajvand and Tarokh (2012) demonstrate that RFM-based segmentation, when combined with CLV modelling, yields significantly more accurate donor prioritisation than either method alone.

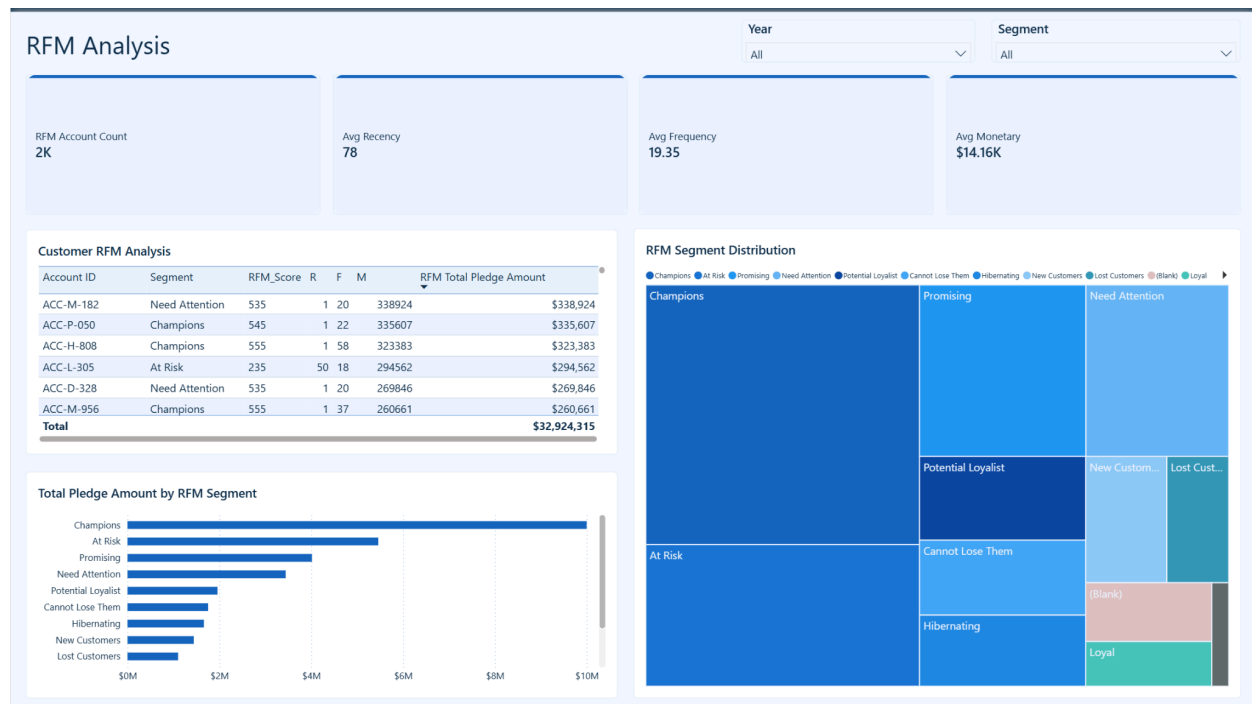


Figure. RFM Analysis page

5. Insight 3: Customer Lifetime Value and Churn Analysis

CLV Distribution

Customer Lifetime Value (CLV) was calculated as the cumulative pledge amount per unique donor across the entire 2019 to 2025 observation window. The average CLV is \$14,132, with a median of \$9,955 and a maximum of \$338,924. The distribution is right-skewed: Very High capacity band donors (581 individuals, 24.9%) contribute \$17.60 million (53.5%), while Low capacity band donors (582 individuals) contribute \$3.50 million (10.6%). This approximately even split across capacity bands suggests that ICMRA's donor acquisition strategy has achieved broad penetration across household income segments.

Churn Assessment

Churn was defined as a donor whose last recorded pledge occurred before 2024. Only 10 donors meet this criterion, yielding a churn rate of 0.4%, an exceptionally low figure by sector standards. Gupta, Lehmann, and Stuart (2004) note that CLV is most valuable as a forward-looking metric when combined with churn probability; ICMRA's negligible observed churn suggests that current CLV figures are sustainable, provided donor engagement levels are maintained. However, the declining revenue trend in 2024 to 2025 raises the possibility that while donors are not churning (they remain active), their per-transaction giving amounts may be decreasing, a phenomenon known as "silent attrition" (Sargeant & Woodliffe, 2007).

Implications for Strategy

The capacity band analysis reveals that the median donor (\$9,955 CLV) sits in the Medium band, while the average (\$14,132) is pulled upward by Very High outliers. Strategies to move Medium donors upward, through targeted upgrade asks, recurring giving programmes, or major gift cultivation, could yield substantial returns. If the 581 Medium-band donors increased their average CLV by just 20% (approximately \$1,750 each), this would represent over \$1 million in incremental revenue.

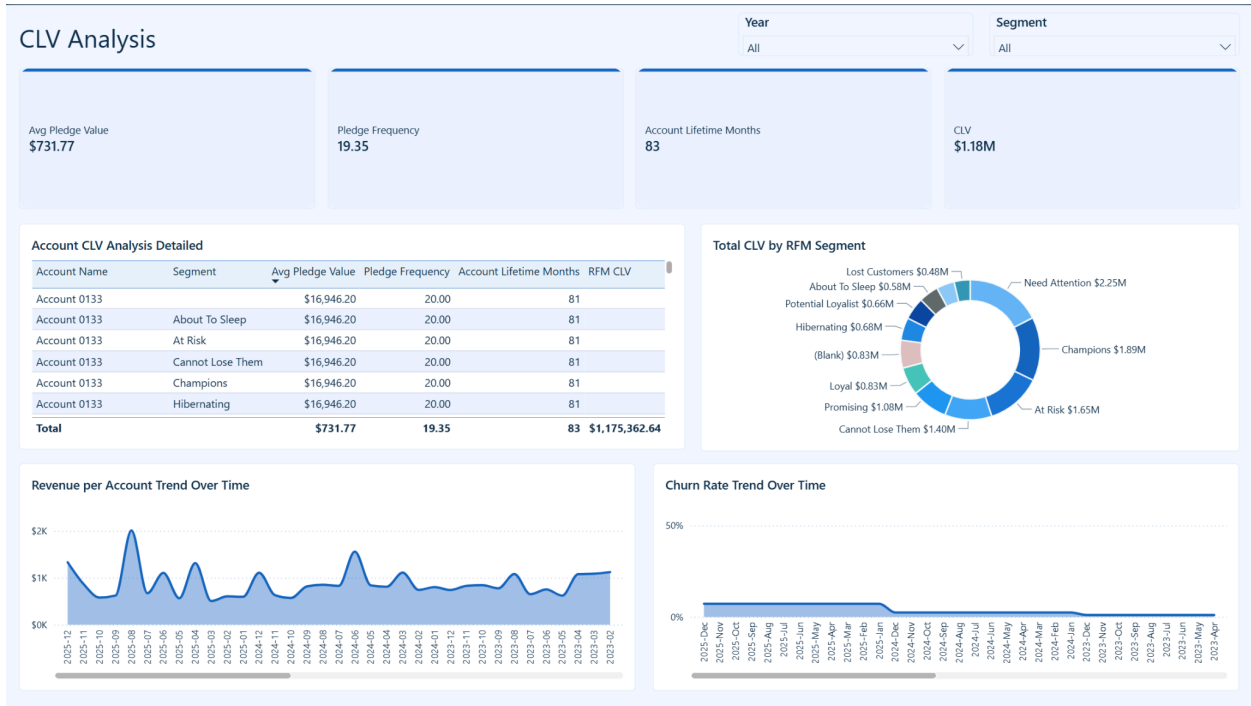


Figure. CLV Analysis

6. Insight 4: Market Basket Analysis: Cross-Campaign Affinity Networks

Methodology

Market Basket Analysis, grounded in association rule mining (Agrawal, Imieliński, & Swami, 1993), was applied to identify co-donation patterns: campaigns that the same donors contribute to simultaneously. For each unique donor, the set of campaigns they have supported was catalogued, and the frequency of every campaign pair co-occurrence was computed to derive support metrics.

Key Findings

The results reveal an extraordinarily high degree of cross-campaign participation. A total of 2,325 donors (99.8%) have contributed to two or more campaigns, with an average of 5.05 campaigns per donor. The strongest campaign pair is the Young Doctor Training Grant Program and the General Oncology Research Grant Program, which share 2,202 donors (94.5% support). This means that nearly all donors who give to one of these campaigns also give to the other. The top six campaign pairs all exhibit support levels above 80%, indicating that ICMRA's donor base treats its major campaigns as a portfolio rather than making isolated contributions.

The practical implication is that cross-selling between campaigns is largely a solved problem; donors already participate broadly. The opportunity instead lies in leveraging these affinity networks to introduce donors to underperforming campaigns. The General Appeal and Grant Appeal campaign categories generated only \$1.09 million and \$1.01 million respectively, with target achievement rates of just 21% and 16%. Redirecting even a small fraction of the established cross-campaign donor traffic toward these underperforming campaigns could significantly improve their results.

Campaign Type Performance

Program Funding campaigns dominate at \$25.97 million in revenue with a 2,310% ROI and 243% target achievement. Flagship Appeals generated \$4.85 million (618% ROI, 61% target achievement). General Appeals and Grant Appeals, while smaller, show ROIs of 126% and 80% respectively, indicating positive but suboptimal returns. The underperformance of these categories relative to their targets (21% and 16% achievement) suggests misaligned target-setting or insufficient promotional investment rather than fundamental unattractiveness to donors.

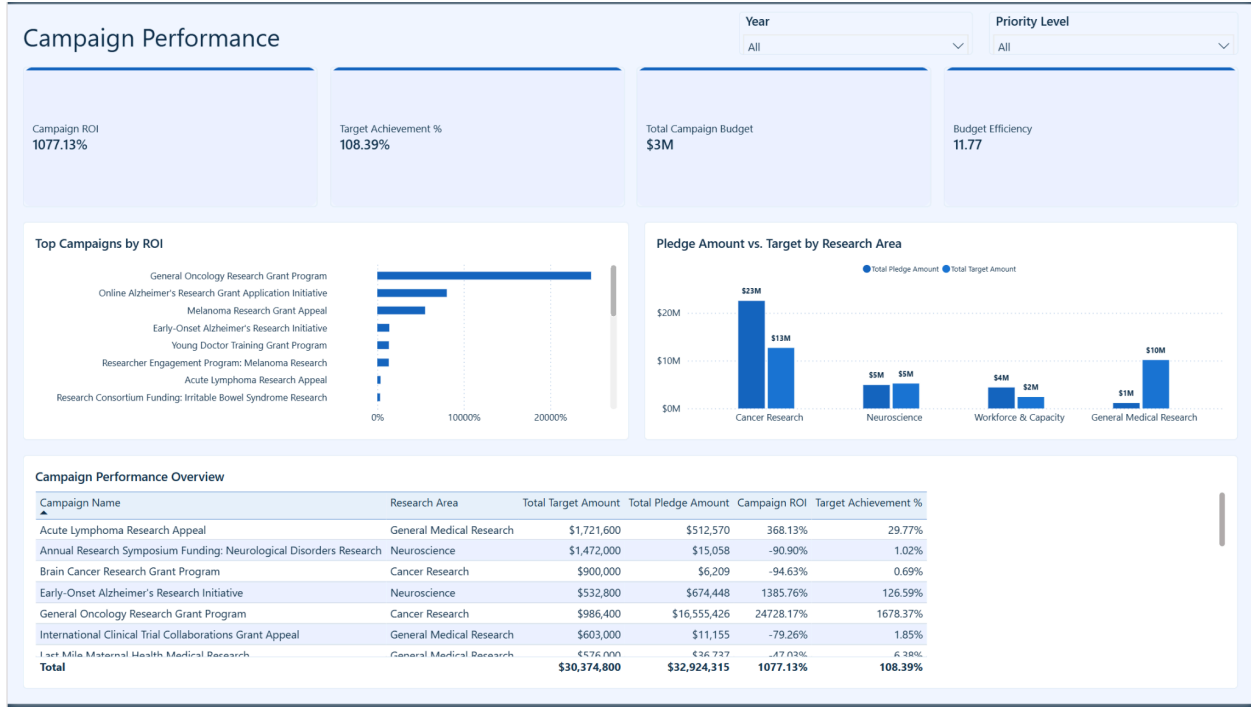


Figure. Campaign Performance Page

7. Insight 5: What-If Scenario Planning and Budget Optimisation

Methodology

A What-If analysis was implemented within the Power BI model using two parameter tables: Budget Adjustment (ranging from -50% to +100% in 5% increments) and Target Adjustment (same range). These parameters dynamically recalculate Adjusted Budget, Adjusted Target, Adjusted ROI, and Adjusted Target Achievement, enabling interactive scenario exploration by the Board.

Scenario Results

Three illustrative scenarios demonstrate the sensitivity of ICMRA's ROI to budget changes. At the baseline (0% adjustment), the total campaign budget of \$2,796,996 against total pledges of \$32,926,665 yields a 1,077% ROI. A 50% budget increase to \$4,195,494 reduces ROI to 685%, while a 50% budget reduction to \$1,398,498 inflates ROI to 2,254%. The break-even budget, the point at which campaign spending equals total pledges, is \$32,926,666, roughly 12 times the current budget.

Budget Efficiency Context

ICMRA's current budget efficiency of \$11.78 raised per dollar spent is exceptional, but this metric must be interpreted cautiously. The total target amount across all campaigns is \$44.96 million, yet only \$32.93 million has been raised, a target achievement rate of 73.2% and a gap of \$12.04 million. This gap suggests that while ICMRA is efficient with the budgets it allocates, the budgets themselves may be undersized relative to the fundraising ambitions encoded in campaign targets. If the Board were to increase total campaign budgets by 50% (to \$4.20 million) while maintaining current efficiency ratios, projected revenue could reach approximately \$49.4 million, exceeding the current total target. Montgomery and Urban (1969) established that scenario modelling of this nature enables organisations to identify the budget allocation that maximises expected return, a principle directly applicable to ICMRA's planning process.

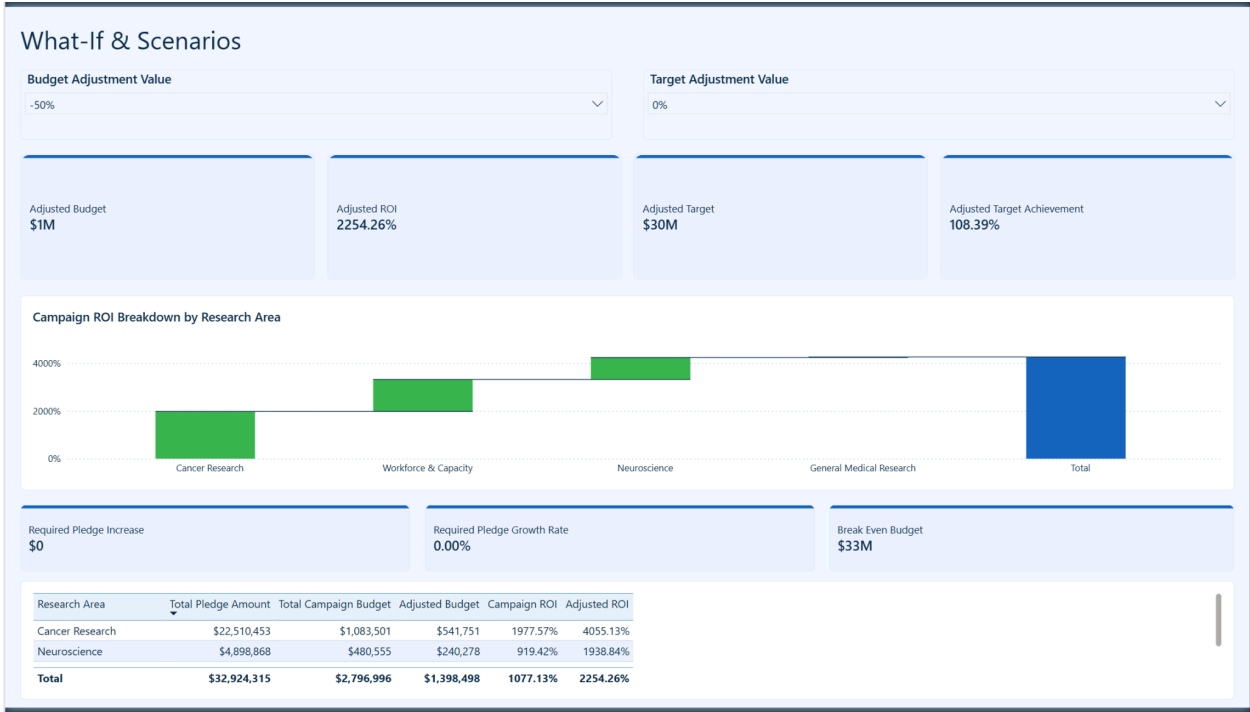


Figure. What-If & Scenarios

8. Insight 6: Cohort Analysis: Longitudinal Donor Value Trajectories

Methodology

Donors were grouped into cohorts based on their year of first recorded pledge (2019, 2020, 2021, 2022, and 2023). For each cohort, the cumulative revenue contribution was tracked across all subsequent years to assess retention patterns and lifetime value trajectories.

Longitudinal Findings

The 2019 cohort demonstrates the strongest longitudinal value. Starting at \$821K in their acquisition year, this cohort's annual contribution grew to \$1,348K by 2021, \$2,977K by 2022, and peaked at \$3,034K in 2024 before declining to \$1,761K in 2025. This represents a 3.7× increase from acquisition year to peak, a pattern consistent with deepening donor engagement over time. The 2020 cohort follows a similar trajectory, growing from \$854K to \$3,232K by 2023 (a 3.8× increase). These patterns confirm that donors acquired in earlier years have substantially higher long-term value than more recently acquired donors.

The 2021 cohort, while smaller, still shows meaningful growth from \$1,014K to \$1,623K by 2023. The 2022 cohort contributed only \$274K in its acquisition year, growing modestly to \$516K by 2024 and then declining to \$160K in 2025. The 2023 cohort is negligible at \$1K. This progressive diminishment of cohort scale is a leading indicator of donor acquisition challenges. Reichheld (1996) demonstrated that cohort-based retention analysis is among the most reliable predictors of long-term organisational health; the declining cohort values at ICMRA signal that investment in new donor acquisition is increasingly urgent.

Seasonality Overlay

Monthly analysis reveals that December is the strongest giving month at 14.4% of annual revenue (\$4.74 million), followed by March (9.5%) and January (9.4%). Q4 dominates at 29.8% of annual revenue. This seasonality should inform cohort acquisition timing: donors acquired during Q4 may exhibit higher initial giving but lower long-term retention, while Q1-acquired donors may show steadier giving patterns. Kumar and Reinartz (2016) argue that cohort analysis, when combined with seasonality adjustments, provides the most accurate basis for CLV forecasting.

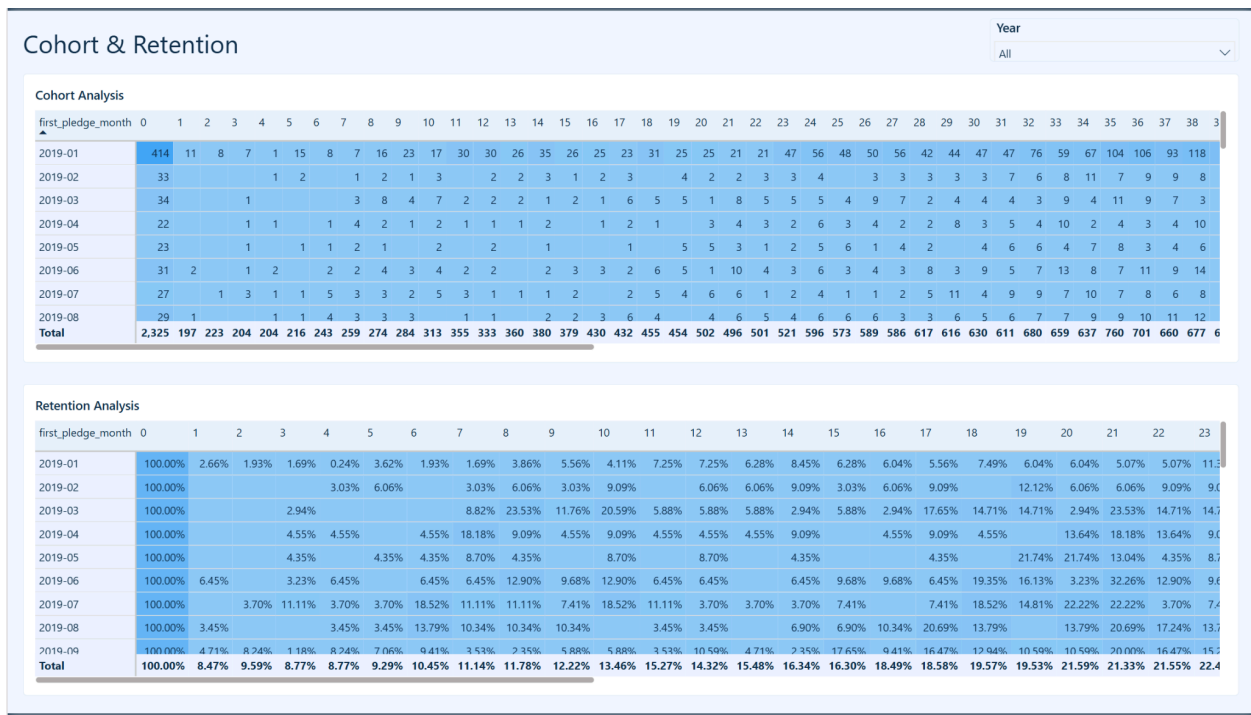


Figure. Cohort & Retention page

9. Strategic Recommendations

Based on the six insights presented, the following recommendations are offered to the ICMRA Board:

1. Revenue Recovery Task Force. Establish a dedicated working group to diagnose the root causes of the 2024 to 2025 revenue decline. The deceleration was already visible in 2023's +8.4% growth (down from +115% in 2022), indicating a structural rather than cyclical shift. Actions should include a review of Cam-Res-App-242's performance and an assessment of donor fatigue or competitive pressures.

2. Geographic Diversification. With 70.7% of revenue from the United States, ICMRA faces material concentration risk. European, Asian, and Oceanian markets represent only 13.9% of revenue. A phased digital fundraising strategy, beginning with pilots in the UK, Australia, and Japan, is recommended.

3. Donor Type Expansion. Reliance on household donors (94.9%) leaves ICMRA exposed. Enterprise partnerships (\$799K), foundation grants (\$784K), and government funding (\$15K) represent untapped channels requiring a dedicated institutional fundraising team.

4. Campaign Portfolio Rebalancing. The dependency on Cam-Res-App-242 (50.3% of revenue) is an unacceptable concentration risk. Investment should target the underperforming General Appeal and Grant Appeal categories (21% and 16% target achievement), leveraging existing cross-campaign affinity networks.

5. Budget Realignment. A 50% budget increase would reduce ROI from 1,077% to 685%, still an exceptional return, while potentially closing the \$12 million target gap. The Board should consider whether maximising efficiency ratios or absolute revenue is the more appropriate objective.

6. Donor Acquisition Investment. Recent cohorts (2022 to 2023) are substantially smaller than earlier ones. Priority should be given to converting the 334 Potential Loyalist donors into Champions and re-engaging the 49 At Risk donors before they enter the Lost segment.

10. References

Agrawal, R., Imieliński, T., & Swami, A. (1993). Mining association rules between sets of items in large databases. *Proceedings of the 1993 ACM SIGMOD International Conference on Management of Data*, 207–216. <https://doi.org/10.1145/170035.170072>

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Sargeant, A., & Woodliffe, L. (2007). Building donor loyalty: The antecedents and role of commitment in the context of charity giving. *Journal of Nonprofit & Public Sector Marketing*, 18(2), 47–68. https://doi.org/10.1300/J054v18n02_03

Appendix A — Data Quality Log

Check	Table	Field(s)	Result	Action
Null check	Medical Research Grant Appeal	All columns	0 nulls in critical fields	None required
Negative values	Medical Research Grant Appeal	Pledge-Pay Amount	0 negative values	None required
Zero values	Medical Research Grant Appeal	Pledge-Pay Amount	0 zero values	None required
Referential integrity	Medical Research Grant Appeal	Account ID, Campaign ID, Cont Type ID	All FKs resolve	None required
Date range validity	Medical Research Grant Appeal	Appeal Close Date, Payment Date	All within 2019–2025	None required
Label standardisation	Account	Account Type	5 records "Househld"	Replaced with "Household"
Campaign count	MRA Campaign	Campaign Code	22 unique campaigns	Consistent with fact table
Contribution type count	Contribution	Contribution_ID	10 unique types	Consistent with fact table

Appendix B — Power Query Transformation Steps

Account Table

1. **Source** — Excel.Workbook connection to ICMRA Dataset 2019–2025.xlsx
2. **Navigation** — Select Account sheet
3. **Promoted Headers** — First row as column headers
4. **Changed Type** — Enforce data types: Account Code (text), Account Name (text), Contact Person (text), Account Type (text), Country (text), Region (text), Account Segment (text), Preferred Referral Source (text), Funding Capacity Band (text), Account Since (datetime), Organisation Size Band (text)
5. **Replaced Value** — Account Type: "Househld" replaced with "Household" for standardisation

Medical Research Grant Appeal (Fact Table)

1. **Source** — Excel.Workbook connection to ICMRA Dataset 2019–2025.xlsx
2. **Navigation** — Select Medical Research Grant Appeal sheet
3. **Promoted Headers** — First row as column headers
4. **Changed Type** — Enforce data types: Pledge ID (text), Account ID (text), Campaign ID (text), Cont Type ID (text), Referral Source (text), Appeal Close Date (datetime), Payment Date (datetime), Pledge-Pay Amount (decimal)

MRA Campaign Table

1. **Source** — Excel.Workbook connection to ICMRA Dataset 2019–2025.xlsx
2. **Navigation** — Select MRA Campaign sheet
3. **Promoted Headers** — First row as column headers
4. **Changed Type** — Enforce data types across all 11 columns (text, datetime, int64)

Contribution Table

1. **Source** — Excel.Workbook connection to ICMRA Dataset 2019–2025.xlsx
2. **Navigation** — Select Contribution sheet
3. **Promoted Headers** — First row as column headers
4. **Changed Type** — Contribution_ID (text), Contribution Type (text), Contribution Group (text)

Dim_Date (Custom Date Table)

- 1. Generated from DAX calculated table using CALENDARAUTO() to span the full date range of the fact table
- 2. Columns added: Date, Year, Quarter, Month, Month Name, Year-Month, Day of Week
- 3. Marked as official date table with relationship to Medical Research Grant Appeal[Appeal Close Date]

Executive Infographic

